

Robust fixed-wing UAV guidance with circulating artificial vector fields

Adriano M. C. Rezende, Vinicius M. Gonçalves, Guilherme V. Raffo, Luciano C. A. Pimenta¹

Abstract—This paper presents a guidance vector field strategy to control a fixed-wing UAV (unmanned aerial vehicle) subject to uncertainty in order to converge to and circulate a closed curve in $\mathbb{R}^3$. The control system is designed based on a reference model of the airplane with constrained input controls. The law is independent of the vector field's structure, however, some analysis considers a consolidated vector field approach. Asymptotic stability is proven with Lyapunov Theory and ultimate bounds are found when bounded uncertainties are taken into account. The control law is continuous except in the surroundings of the unavoidable field's singularities. A theorem ensures asymptotic convergence when a switch is made. Simulations with a 6 DOF, 12 states realistic aircraft model demonstrate the efficiency of the strategy and its advantages.

I. INTRODUCTION

Nowadays UAVs are used in many military and civil applications. In particular, fixed-wing vehicles are used in tasks such as surveillance, monitoring, convoy protection, and others. Fixed-wing UAVs have an important advantage regarding to autonomy in terms of energy in comparison to rotating wing vehicles. In this context, guidance techniques for such aircrafts are receiving a lot of attention in the academy and industry.

An extensive survey about several approaches to the guidance problem is presented in [1], whose destination to practitioners reaffirms the importance of such methods. For instance, line-of-sight (LOS) [2] and nonlinear guidance laws (NLGL) [3] are based on a virtual target point, which guides the UAV towards the path. A different approach, which is simple and generates good trajectories, considers a guidance vector field (GVF) [4] together with a control law to make the UAV converge to a desired path. In the present work we develop a controller that makes a UAV to converge to the integral lines of a generic vector field. In other words, we mean that our method is not restricted to straight lines and circular loiters as usual (see [1]). In addition, different from all works in the survey [1], where the paths have a fixed height, our method can be applied to non planar curves.

Different approaches to the fixed-wing UAV control includes the use of Model Predictive Control (MPC) for path tracking. In [5] an MPC is developed to enforce a UAV

track a path in 2D. In fact, the limitation of fixed height is very common in the literature. In [6] an MPC combined with $\mathcal{L}_1$ -navigation strategy is used to make a fixed-wing UAV track a desired path in 3D. In [7] quaternion algebra is applied to perform the path tracking in 3 dimensions, where straight lines and planar circles are considered. As previously mentioned our proposed strategy is also able to deal with non planar curves described in 3D space.

Guidance vector field laws are many times addressed with straight lines and circles [8]. In [9], GVF laws initially developed for circles are then applied to different shapes obtained by covariant transformations. In [10] a controller is developed for generic curve shapes. Their method is similar to ours, since the UAV yaw angle controller, which makes the UAV orient itself with the field, is composed of a feed forward term and a proportional one. They point out the difficulties in finding a formal proof establishing that getting oriented asymptotically with the field leads to convergence to the path, despite this convergence happens in all practical examples. Here we present a formal convergence proof for 3D cases. In [11], controllers to guide a UAV in 3 dimensional vector fields are developed. Convergence of the two proposed fields are presented, but no proof is presented regarding the guidance controllers.

Obtaining convergence proofs for the proposed controller is an important objective of this work. In [12] a simplified non-holonomic model is considered in order to prove the convergence to a circular path with a hybrid controller. Despite the use of the simplified model they use a set of low level controllers that impose the simplified dynamics to the actual UAV. In [13] the work is extended with uncertainties in the model, which may account for modeling errors and external disturbances. In the presence of uncertainties the UAV converges to a band around the target circle. The present work extends the results of [13] considering uncertainties not only for circles but also for generic curves.

Although our controllers are directly applicable to any smooth vector field, such as the ones generated with Fourier basis curves [14], we focus our analysis on the fields proposed in [15]. This method considers closed paths in 3D defined as the intersection of level sets of two smooth scalar functions.

Our method is able to deal with the majority of the details commonly addressed separately in different works that consider GVF, recalling: 3 dimensional space [11]; generic curve shapes [10], [11], being independent of the vector field construction; and model uncertainties [13]. In addition, we formally treat the problem of singularities of the vector field. We prove a theorem establishing sufficient

¹ The authors are with the Graduate Program in Electrical Engineering of the Federal University of Minas Gerais (UFMG). E-mail: adrianomcr18@gmail.com, mariano@cpdee.ufmg.br, raffo@ufmg.br, lucpim@cpdee.ufmg.br

*The authors gratefully acknowledge the financial support of the Brazilian agencies CNPq, CAPES, FAPEMIG and National Institute of Science and Technology for Cooperative Autonomous Systems Applied to Security and Environment (InSAC). The authors also acknowledge José Lucas Gomes Olavo for the help with the simulations with the realistic aircraft model.

conditions to state that if we switch the control law around these singularities we still ensure convergence to the desired path. This is in contrast to [10], which provides a set of initial conditions that lead to convergence to the path without passing through the singularities. Thus, our contribution is a guidance vector field technique that considers generic curve shapes in 3D while dealing with uncertainties in the aircraft reference model and singularities in the field.

The paper is organized as follows: In the next section the control problem is formally stated. In section III the vector field strategy proposed in [15] is reviewed. The control laws are developed in Section IV. In Section V the strategy to deal with singularities is presented and the closed-loop stability is proved. Realistic simulation results are shown in Section VI and conclusions and future works presented in Section VII.

II. PROBLEM STATEMENT

The main problem of this work is to establish a control law for a fixed-wing UAV such that it converges to and circulates a closed curve in a given direction. We consider that this closed curve, namely $\mathcal{C}$, is the limit cycle of an artificial vector field defined in $\mathbb{R}^2$ or $\mathbb{R}^3$ space. This vector field should guide the reference of the controller.

As addressed in [12] and [13], a reference model is considered for the fixed-wing UAV. This simplified model is valid since it is used in a cascade scheme with a lower level controller. The model is described by

$$\dot{x} = v \cos(\theta), \quad (1a)$$

$$\dot{y} = v \sin(\theta), \quad (1b)$$

$$\dot{z} = \tau_z^{-1}(-z + z_c) + u_z, \quad (1c)$$

$$\dot{\theta} = \tau_\theta^{-1}(-\theta + \theta_c) + u_\theta, \quad (1d)$$

$$\dot{v} = \tau_v^{-1}(-v + v_c) + u_v, \quad (1e)$$

where x , y and z are the Cartesian coordinates of the center of mass of the UAV, θ is the yaw angle, and v is the forward speed in the xy plane with respect to the atmosphere. The control inputs of altitude, yaw angle and forward velocity are z_c , θ_c and v_c , respectively. The time constants, all strictly positive, associated to z , θ , and v are τ_z , τ_θ and τ_v , respectively. The terms u_z , u_θ and u_v are additive uncertainties with absolute value limited by U_z , U_θ and U_v , respectively, i.e.

$$|u_z| \leq U_z, \quad |u_\theta| \leq U_\theta, \quad |u_v| \leq U_v. \quad (2)$$

Uncertainties may account for differences between the simplified model (1) as well for measurement and actuation errors. Besides, the following constraints are added to (1)

$$|z_c - z| \leq \tau_z v_z^{max}, \quad (3a)$$

$$|\theta_c - \theta| \leq \tau_\theta \omega^{max}, \quad (3b)$$

$$v^{min} \leq v_c \leq v^{max}, \quad (3c)$$

in which v_z^{max} is the maximum absolute value of the velocity in z , ω^{max} is the maximum turning rate, and $v^{min} > 0$ and v^{max} are the minimum and maximum forward speeds. Note

that $v^{min} > 0$, which represents the necessity of the airplane to maintain its aerodynamic lift.

Although the model (1) is a simplified representation of an aircraft, the constraints (3) make it a good approximation of UAV dynamics. In [13] some simulations show the accuracy of this model. Therefore, one should have in mind that a set of lower level controllers are used to impose the desired dynamics of the reference model (1).

The main problem addressed in this work can be stated as follows:

Problem 1: Find a set of control laws, for z_c , θ_c and v_c , which obey the actuation constraints (3) and make the fixed-wing UAV, subject to bounded uncertainties, described by (1) and (2), converge to an integral line of a given guidance vector field.

III. GUIDANCE VECTOR FIELD REVIEW

The control laws developed in the next section can be applied to any smooth vector field in 3 dimensions, including those with open integral lines. However, we focus some of our further analysis (Section IV-D) in the vector field structure proposed in [15]. There the authors propose a strategy for generating vector fields to converge to and circulate a closed time varying curve $\mathcal{C}$ in n -dimensional spaces. Here we aim to rapidly recall the vector field strategy. Let n be the dimension of the workspace and $\mathbf{p} \in \mathbb{R}^n$ a coordinate for this space. The strategy in [15] relies on the ability of finding $n - 1$ well-behaved scalar functions $\alpha_1(\mathbf{p})$, $\alpha_2(\mathbf{p})$, ... $\alpha_{n-1}(\mathbf{p})$ such that the intersection of the level sets $\alpha_i(\mathbf{p}) = 0$, $i = 1, 2, \dots, n - 1$, implicit defines the target curve $\mathcal{C}$. In the present work we do not tackle the problem of finding these functions. Next, the structures of the vector fields for 2 and 3 dimensions are presented.

Let $\Phi(\mathbf{p}) : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be the vector field. It will be assumed that n may be 2 or 3, depending on the context. For a field in $\mathbb{R}^2$, we have

$$\Phi(x, y) = G(P) \frac{\nabla P}{\|\nabla P\|} + H(P) \frac{R \nabla \alpha_1}{\|R \nabla \alpha_1\|}, \quad (4)$$

where $R \in SO(2)$ is a rotation matrix of 90° and the operator ∇ performs the gradient with respect to $\mathbf{p}$. The scalar function $P \equiv P(\alpha_1, \dots, \alpha_{n-1}) : \mathbb{R}^{n-1} \rightarrow \mathbb{R}^+$ is zero at $\mathcal{C}$ and positive elsewhere. The function P represents a distance to $\mathcal{C}$, and $P(\mathbf{p}) = 0 \iff \mathbf{p} \in \mathcal{C}$. We also consider a scalar function $G(P) : \mathbb{R}^+ \rightarrow \mathbb{R}^-$, that is $G(P) \leq 0$, such that $G(P) = 0 \iff P = 0$, $|G(P)| \leq 1$. Also, $G(P)$ is decreasing, therefore invertible. The function $H(P)$ is defined as $H(P) = \sqrt{1 - G(P)^2}$. In $\mathbb{R}^3$ we have

$$\Phi(x, y, z) = G(P) \frac{\nabla P}{\|\nabla P\|} + H(P) \frac{\nabla \alpha_1 \times \nabla \alpha_2}{\|\nabla \alpha_1 \times \nabla \alpha_2\|}, \quad (5)$$

where the operator $\times$ corresponds to the cross product.

Note that if $\|\nabla P\| = 0$ or $\|\nabla \alpha_1 \times \nabla \alpha_2\| = 0$ there is a problem with the normalization. We have $\nabla P = 0$ when $\mathbf{p} \in \mathcal{C}$ or in some isolated equilibrium point outside $\mathcal{C}$. The former is not a problem since in this case $G(P) \rightarrow 0$ as well. Besides, we assume the latter are repulsive points. We will assume the same repulsiveness when $\|\nabla \alpha_1 \times \nabla \alpha_2\| = 0$.

Figure 1 illustrates the vector field given by (5), in which the zero level sets of two scalar functions α_1 and α_2 are highlighted in blue and red. The intersection between them (black line) defines the closed curve to which the integral lines of the field converge.

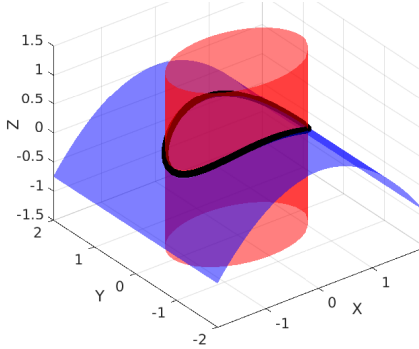


Fig. 1. Two level sets in $\mathbb{R}^3$ and the intersection between them.

The terms in (4) and (5) can be easily interpreted as a convergent component and a circulation component. Let Φ_v be the term that multiplies $G(P)$ and Φ_t the one that multiplies $H(P)$. The components Φ_v and Φ_t are responsible for the convergence to the curve and for the curve tracking, respectively. In [15] some important properties of the field are made, from which two of them are recalled.

Property 1: Considering the simple integrator model, that is, $\dot{\mathbf{p}} = \Phi$, the function $P(\mathbf{p})$ is a Lyapunov function and $\dot{P} = \nabla P^T \Phi = G(P) \|\nabla P\| \leq 0$.

Property 2: For every non fixed point, the convergent component (Φ_v) and the circulating one (Φ_t) are perpendicular to each other, i.e., $\Phi_v \perp \Phi_t$.

Through the development of our controller we are interested only on the orientation of the vector fields. Then we consider the normalized field. Note that convergence properties are not affected by any positive normalization. Let $\hat{\Phi} \equiv [\hat{\Phi}_x \ \hat{\Phi}_y \ 0]^T \equiv \hat{\Phi}(x, y, z) \in \mathbb{R}^3$ be a horizontal field (no component in z) normalized with the xy components of Φ . Let also $\hat{\Phi}_z \equiv \hat{\Phi}_z(x, y, z) \in \mathbb{R}$ be the vertical component of Φ scaled accordingly. Formally, we define

$$\hat{\Phi} = \frac{1}{\sqrt{\Phi_x^2 + \Phi_y^2}} \begin{bmatrix} \Phi_x \\ \Phi_y \\ 0 \end{bmatrix}, \quad \hat{\Phi}_z = \frac{\Phi_z}{\sqrt{\Phi_x^2 + \Phi_y^2}} \quad (6)$$

The definition of the *normalized and projected field* $\hat{\Phi}$ will be very important to control the angle θ . Its necessity is intuitive since θ is a rotation around z world axis, being independent of z component of Φ .

IV. CONTROL

The key idea to control the model (1) is to orient the UAV heading angle with the vector field. Intuitively, by orienting the UAV its non-holonomic constraint is respected, and it will be able to follow the field's integral lines. Thus, we first develop a control law for the turning ratio $\omega(t) \equiv \dot{\theta}$, when $u_\theta = 0$, which is able to orient the UAV with the field. In the sequence, control laws for forward speed and vertical motion are defined. Next we perform an analysis when the disturbances are present.

A. Heading control

Let $\theta_f(\mathbf{p})$ be the orientation of the vector field at point $\mathbf{p}$, i.e., $\theta_f \equiv \text{atan2}(\Phi_y, \Phi_x)$, where atan2 is the four quadrant arc tangent function. This definition allows to compute the robot's orientation error $\bar{\theta} \equiv \bar{\theta}(\mathbf{p}, \theta)$ as $\bar{\theta} = \theta_f - \theta$. Disregard for now the uncertainty u_θ and consider the time-derivative of the error $\bar{\theta}$, which is

$$\dot{\bar{\theta}}(t) = \dot{\theta}_f(x, y, z) - \omega(t). \quad (7)$$

If we are able to stabilize $\bar{\theta}$ at the origin, the UAV will be oriented with the field and will follow its integral lines. The proof that the stabilization of $\bar{\theta}$ leads to convergence of $\mathbf{p}$ to $\mathcal{C}$ will be performed in Subsection IV-D. Now, consider the following Lyapunov candidate function:

$$V_\theta(\bar{\theta}) = 1 - \cos(\bar{\theta}), \quad -\pi < \bar{\theta} < \pi. \quad (8)$$

Note that $V_\theta(0) = 0$ and $V_\theta(\bar{\theta}) > 0 \ \forall \ \bar{\theta} \neq 0$. The time-derivative of (8) is given by

$$\dot{V}_\theta(\bar{\theta}) = \sin(\bar{\theta}) \dot{\bar{\theta}} = \sin(\bar{\theta}) \left[\dot{\theta}_f(x, y, z) - \omega \right]. \quad (9)$$

In order to define a stabilizing control law ω , we need to compute $\dot{\theta}_f(\mathbf{p})$. Then, consider the following theorem.

Theorem 1: Let $\mathbf{p}(t) = [x(t) \ y(t) \ z(t)]^T \in \mathbb{R}^3$ be an arbitrary trajectory, and $\Phi(\mathbf{p}) : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a vector field such that $\Phi(\mathbf{p}) = [\Phi_x(\mathbf{p}) \ \Phi_y(\mathbf{p}) \ \Phi_z(\mathbf{p})]^T$. Consider Φ_x , Φ_y and Φ_z continuous and differentiable functions with respect to x , y and z . Assuming $\Phi_x(t) \equiv \Phi_x(\mathbf{p}(t))$ and $\Phi_y(t) \equiv \Phi_y(\mathbf{p}(t))$, we define the function $\theta_f(t) \equiv \text{atan2}(\Phi_y(t), \Phi_x(t))$. Then, the time-derivative $\dot{\theta}_f(t)$ throughout the trajectory $\mathbf{p}(t)$ is given by

$$\dot{\theta}_f(t) = \left(\hat{\Phi}_x \nabla \hat{\Phi}_y^T - \hat{\Phi}_y \nabla \hat{\Phi}_x^T \right) \dot{\mathbf{p}}, \quad (10)$$

in which $\hat{\Phi} = [\Phi_x \ \Phi_y \ 0]^T (\Phi_x^2 + \Phi_y^2)^{-0.5}$ is the field projected onto the local horizontal plane and normalized. The vectors $\nabla \hat{\Phi}_x$ and $\nabla \hat{\Phi}_y$ are the gradients of $\hat{\Phi}_x$ and $\hat{\Phi}_y$, respectively, with respect to x , y and z .

Proof: First, note that the function θ_f can be computed from $\hat{\Phi}$ in (6). In order to verify this property it is enough to observe that any normalization factor cancels inside the atan function. Then the time-derivative of θ_f can be computed based on $\hat{\Phi}$ as

$$\dot{\theta}_f(t) = \frac{d}{dt} \text{atan} \left(\frac{\hat{\Phi}_y}{\hat{\Phi}_x} \right) = \frac{\dot{\hat{\Phi}}_y \hat{\Phi}_x - \hat{\Phi}_y \dot{\hat{\Phi}}_x}{\hat{\Phi}_x^2 + \hat{\Phi}_y^2}. \quad (11)$$

Note that $\hat{\Phi}_x^2 + \hat{\Phi}_y^2 = 1$. Replacing it in (11) and applying the chain rule to compute the time-derivatives, we obtain

$$\dot{\theta}_f(t) = \left(\nabla \hat{\Phi}_y^T \dot{\mathbf{p}} \right) \hat{\Phi}_x - \left(\nabla \hat{\Phi}_x^T \dot{\mathbf{p}} \right) \hat{\Phi}_y. \quad (12)$$

By isolating $\dot{\mathbf{p}}$, equation (12) can be written as in (10). This concludes the proof. ■

Note that Theorem 1 provides the value of $\dot{\theta}_f$ for any smooth vector field in $\mathbb{R}^3$ and for any direction the UAV is moving. This result enables our strategy to be applicable to 3

dimensional paths. Now, replacing (10) into (9) with $k_p > 0$, the control law is given by

$$\omega(t) = \left(\hat{\Phi}_x \nabla \hat{\Phi}_y^T - \hat{\Phi}_y \nabla \hat{\Phi}_x^T \right) \dot{\mathbf{p}} + k_p \sin(\bar{\theta}), \quad (13)$$

which yields, given $u_\theta = 0$,

$$\dot{V}_\theta(\bar{\theta}) = -k_p \sin(\bar{\theta})^2. \quad (14)$$

Since for $-\pi < \bar{\theta} < \pi$ the function $\dot{V}_\theta(\bar{\theta}) < 0 \forall \bar{\theta} \neq 0$, (8) is indeed a Lyapunov function, and the control law in (13) ensures asymptotically stability of the robot's orientation error at the origin.

It is important to emphasize that for $\bar{\theta} = \pi \sim -\pi$ the function $\dot{V}_\theta(\bar{\theta}) = 0$. In fact, the system admits an equilibrium point at $\bar{\theta} = \pi$. However, this equilibrium point is unstable, since $V_\theta(\bar{\theta})$ has its maximum value at $\bar{\theta} = \pi$ and $\dot{V}_\theta(\pi - \epsilon) < 0$ for any ϵ arbitrarily small. Since $\bar{\theta} \in \mathbb{S}^1$, it is impossible to establish a continuous control law that stabilizes $\bar{\theta}$ at 0, for which there is no other equilibrium point at any $\bar{\theta} \neq 0$.

Let us now define limits for $\omega(t)$ given maximum values of the field derivatives. Consider a positive scalar M such that $\|\nabla \hat{\Phi}_x\| \leq M$ and $\|\nabla \hat{\Phi}_y\| \leq M$. Note that in (13), $\hat{\Phi}_x = \cos(\theta_f)$ and $\hat{\Phi}_y = \sin(\theta_f)$. Then, by using triangular inequality in (13) we can find

$$|\omega(t)| \leq |\cos(\theta_f) - \sin(\theta_f)|M\|\dot{\mathbf{p}}\| + |k_p \sin(\bar{\theta})|. \quad (15)$$

The maximum $\mathcal{L}_2$ -norm of the difference of trigonometric functions is $\sqrt{2}$, thus

$$|\omega(t)| \leq \sqrt{2}M\|\dot{\mathbf{p}}\| + k_p. \quad (16)$$

Then, in order to obey (3b) it is sufficient that:

$$M \leq \frac{\omega^{max} - k_p}{\sqrt{2}\|\dot{\mathbf{p}}\|^{max}}, \quad (17)$$

where $\|\dot{\mathbf{p}}\|^{max}$ is the maximum speed the UAV may reach. Note that to control θ , it is required that the vector field itself obey condition (17) in order to (3b) be satisfied.

Finally, given equations (1d) and (13) it is possible to define the control θ_c as

$$\theta_c = \tau_\theta \left(\hat{\Phi}_x \nabla \hat{\Phi}_y^T - \hat{\Phi}_y \nabla \hat{\Phi}_x^T \right) \dot{\mathbf{p}} + \tau_\theta k_p \sin(\bar{\theta}) + \theta. \quad (18)$$

B. Velocity control

Note that the control law ω , defined in (13), considers the velocity $v(t)$ in $\dot{\mathbf{p}}$. This allows a decay of the Lyapunov function V_θ that is not dependent on v , (see (14)). This decoupling between $v(t)$ and $\omega(t)$ allows the definition of independent references for $v(t)$. For simplicity, here we will choose a constant forward speed v_r to the UAV. Since the system (1e) is already stable, we define

$$v_c = v_r, \quad v^{min} \leq v_r \leq v^{max}. \quad (19)$$

C. Altitude control

The dynamics of the height, z , in model (1) is completely decoupled from others. In (6) we have defined $\hat{\Phi}_z$ accordingly to the normalization factor $\|[\hat{\Phi}_x \ \hat{\Phi}_y]\|$. Then, in order to make the UAV follow the field's integral lines, we need to scale a desired velocity $\tilde{V}_z$ to $\hat{\Phi}_z$ with the same factor that

$\|[\dot{x} \ \dot{y}]\|$ is scaled to $\|[\hat{\Phi}_x \ \hat{\Phi}_y]\|$. Since $\|[\hat{\Phi}_x \ \hat{\Phi}_y]\| = 1$, we have

$$\tilde{V}_z = \|[\dot{x} \ \dot{y}]\| \hat{\Phi}_z. \quad (20)$$

The computation in (20) may still result in $|\tilde{V}_z| > v_z^{max}$. Then, the following saturation function is defined

$$V_z = \begin{cases} \tilde{V}_z, & \text{if } |\tilde{V}_z| \leq v_z^{max}, \\ v_z^{max} \tilde{V}_z / |\tilde{V}_z|, & \text{if } |\tilde{V}_z| > v_z^{max}. \end{cases} \quad (21)$$

Note that assuming feasibility for the path $\mathcal{C}$ we ensure that the saturation in (21) is inactive at $\mathcal{C}$. Also, a formal convergence proof is still necessary when the saturation is considered, even that, in practice, the convergence is always reached. Finally, according to (1c) the height control is defined as

$$z_c = \tau_z V_z + z. \quad (22)$$

D. Disturbance analysis

If we consider the disturbance u_θ , and assume the control law (13), the time-derivative of the Lyapunov function (14) becomes

$$\dot{V}_\theta = -k_p \sin(\bar{\theta})^2 - \sin(\bar{\theta})u_\theta. \quad (23)$$

Considering the worst scenario ($|u_\theta| = U_\theta$), and by using the triangular inequality, it is possible to write

$$\dot{V}_\theta \leq |\sin(\bar{\theta})| (-k_p |\sin(\bar{\theta})| + U_\theta). \quad (24)$$

From (24), we see that $|\sin(\bar{\theta})| > U_\theta k_p^{-1} \implies \dot{V}_\theta < 0$. Then, defining $\gamma = \text{asin}(U_\theta k_p^{-1})$, we have the following invariant set

$$\mathcal{S}_\theta = \{\bar{\theta} \in \mathbb{S} \mid |\bar{\theta}| \leq \gamma\}. \quad (25)$$

Equation (25) establishes an ultimate bound for the error on the state θ . Note that in order to this ultimate bound exist it is necessary that $k_p > U_\theta$. It means that we can not ensure convergence if the ability to correct errors, k_p , is smaller than the uncertainty, U_θ . Also, it is important to mention that $\dot{V}_\theta$ may be positive for some values of u_θ when $\pi - |\bar{\theta}| < \gamma$. Thus, rigorously, convergence of θ to the origin requires $|\bar{\theta}(0)| < \pi - \gamma$. In practice, convergence happens with any initial condition $|\bar{\theta}(0)|$, since $\pi - |\bar{\theta}| < \gamma$ does not define an invariant set.

It is still necessary to analyze if the ultimate boundedness on $\bar{\theta}$ implies an ultimate boundedness on the function P (potential function that is zero at the target curve $\mathcal{C}$). This is verified if a limited error $\bar{\theta}$ implies in a limited distance from the target curve. Now, consider the following Lemma.

Lemma 1: Let $w_1, w_2, w_3 \in \mathbb{R}^3$ vectors that respect

$$w_1^T w_3 = A, \quad (26a)$$

$$w_2^T w_3 \geq B, \quad (26b)$$

$$\|w_1\| = \|w_2\| = \|w_3\| = 1, \quad (26c)$$

in which $-1 \leq A \leq 0$ and $0 \leq B \leq 1$ are constants. Then

$$w_1^T w_2 \leq AB + \sqrt{1 - A^2} \sqrt{1 - B^2}. \quad (27)$$

Proof: Let $r = w_1^T w_2$. Consider a matrix $Q = w_1 w_1^T - w_2 w_2^T$. One can show that the eigenvalues of Q are 0, $+\sqrt{1-r^2}$ and $-\sqrt{1-r^2}$. Then

$$\|Q w_3\|^2 \leq \|Q\|^2 \|w_3\|^2 = \|Q\|^2. \quad (28)$$

We have that $\|Q\| = \sqrt{1-r^2}$ because Q is symmetric and hence its Euclidean induced norm is the greatest eigenvalue in absolute value. On the other hand, from $\|Q w_3\|^2 = \|(w_1 w_1^T - w_2 w_2^T) w_3\|^2 = (w_1^T w_3)^2 + (w_2^T w_3)^2 - 2(w_1^T w_3)(w_2^T w_3)r$, and using (28) we have

$$(w_1^T w_3)^2 + (w_2^T w_3)^2 - 2(w_1^T w_3)(w_2^T w_3)r \leq 1 - r^2. \quad (29)$$

Defining $C = w_2^T w_3$ and using $w_1^T w_3 = A$, we have

$$r^2 - 2ACr + A^2 + C^2 - 1 \leq 0. \quad (30)$$

If we solve the quadratic inequality in (30), we obtain

$$r \leq AC + \sqrt{1-A^2}\sqrt{1-C^2}. \quad (31)$$

We also have $A \leq 0$ and $C \geq B \geq 0$, $AC \leq AB$ and $\sqrt{1-C^2} \leq \sqrt{1-B^2}$. Applying these facts to (31) we prove (27). ■

Remark 1: For functions $f_1(r)$, $f_2(r)$, $g_1(r)$ and $g_2(r) > 0$ and a set $\mathcal{R}$ we have that:

$$\min_{r \in \mathcal{R}} (f_1(r) + f_2(r)) \geq \min_{r \in \mathcal{R}} f_1(r) + \min_{r \in \mathcal{R}} f_2(r). \quad (32)$$

$$\min_{r \in \mathcal{R}} \frac{g_1(r)}{g_2(r)} \geq \frac{\min_{r \in \mathcal{R}} g_1(r)}{\max_{r \in \mathcal{R}} g_2(r)}. \quad (33)$$

Now, consider a vector field Φ as defined in (5). Recalling that $\dot{P} = \nabla P^T \dot{\mathbf{p}}$, we can rewrite it as

$$\dot{P} = \|\nabla P\| \|\dot{\mathbf{p}}\| \left(\frac{\nabla P}{\|\nabla P\|} \right)^T \frac{\dot{\mathbf{p}}}{\|\dot{\mathbf{p}}\|}. \quad (34)$$

Considering, without loss of generality, that $v = \sqrt{\Phi_x^2 + \Phi_y^2}$, and $u'_z = u_z \sqrt{\Phi_x^2 + \Phi_y^2}/v_c$ we can write the vector with disturbance:

$$\dot{\mathbf{p}} = \begin{bmatrix} \sqrt{\Phi_x^2 + \Phi_y^2} \cos(\theta) \\ \sqrt{\Phi_x^2 + \Phi_y^2} \sin(\theta) \\ \Phi_z + u'_z \end{bmatrix}. \quad (35)$$

We assume that $|\bar{\theta}| \leq \gamma$ and $|u'_z| \leq \zeta$. Since $\dot{\mathbf{p}}$ in (35) is scaled to v_c we need to define $\zeta = U_z/v_c$. We will apply Lemma 1 to conclude our main result of this section, but first we need to define

$$w_1 = \frac{\nabla P}{\|\nabla P\|}, \quad w_2 = \frac{\dot{\mathbf{p}}}{\|\dot{\mathbf{p}}\|}, \quad w_3 = \Phi, \quad (36)$$

and then compute the respective values in Lemma 1. Recalling Property 2 it is straightforward that:

$$w_1^T w_3 = G(P) \leq 0. \quad (37)$$

The vector w_3 can be written as (35) with $\theta = \theta_f$ and $u'_z = 0$. From (35), we have that $\|\dot{\mathbf{p}}\| = \sqrt{1 + 2\Phi_z u'_z + u'^2_z}$. Then, separating the xy components of w_2 and w_3 from the

z components, and recalling $\bar{\theta} = \theta_f - \theta$, we can compute $w_2^T w_3$ as follows

$$w_2^T w_3 = \frac{(\Phi_x^2 + \Phi_y^2) \cos(\bar{\theta}) + \Phi_z^2 + \Phi_z u'_z}{\sqrt{1 + 2\Phi_z u'_z + u'^2_z}}. \quad (38)$$

From now, we will successively lower bound $w_2^T w_3$ by using bounds on the variables $\bar{\theta}$, Φ_z , and finally on u'_z . Beginning with $|\bar{\theta}| \leq \gamma \leq \pi/2$, we have that $\cos(\bar{\theta}) \geq \cos(\gamma)$ and $\Phi_x^2 + \Phi_y^2 = 1 - \Phi_z^2 \geq 0$. Thus

$$w_2^T w_3 \geq \frac{(1 - \Phi_z^2) \cos(\gamma) + \Phi_z^2 + \Phi_z u'_z}{\sqrt{1 + 2\Phi_z u'_z + u'^2_z}}. \quad (39)$$

Now, we will lower bound the right hand side of (39) on $|\Phi_z| \leq 1$. In this case, consider $r = \Phi_z$, $\mathcal{R} = \{r \mid |r| \leq 1\}$ and $f_1(r) = (1 - r^2) \cos(\gamma) + r^2$, $f_2(r) = r u'_z$, $g_1(r) = f_1(r) + f_2(r)$, and $g_2(r) = \sqrt{1 + 2r u'_z + u'^2_z} > 0$. Now, applying equations (33) and (32), we conclude that

$$w_2^T w_3 \geq \frac{\cos(\gamma) - |u'_z|}{\sqrt{1 + 2|u'_z| + u'^2_z}} = \frac{\cos(\gamma) - |u'_z|}{1 + |u'_z|}. \quad (40)$$

Finally, we will lower bound the right hand side of (40) on $|u'_z| \leq \zeta$. For that, consider $r = u'_z$, $\mathcal{R} = \{r \mid |r| \leq \zeta\}$ and the functions $g_1(r) = \cos(\gamma) - |r|$ and $g_2(r) = 1 + |r| > 0$. Then, applying (33):

$$w_2^T w_3 \geq \frac{\cos(\gamma) - \zeta}{1 + \zeta} = B. \quad (41)$$

Now we can state the following theorem:

Theorem 2: Let $\gamma \leq \pi/2$ and $\zeta \leq \cos(\gamma)$. Then, there exists $\chi \geq 0$ such that $\cos(\chi) = B$ and we can prove that the set $\mathcal{S} = \{\mathbf{p} \mid P(\mathbf{p}) \leq G^{-1}(-\sin(\chi))\}$ is positive invariant.

Proof: First, note that $A = G(P) \leq 0$ and since $\zeta \leq \cos(\gamma) \leq 1$, then $0 \leq B \leq 1$. Therefore, there exists $\chi \geq 0$ such that $\cos(\chi) = B$. Now, defining $D = \|\nabla P\| \|\dot{\mathbf{p}}\| \geq 0$ and recalling equations (34), (36), (37) and (40) we are in position to apply Lemma 1 and conclude that.

$$\dot{P} \leq D \left(AB + \sqrt{1 - A^2} \sqrt{1 - B^2} \right). \quad (42)$$

Let $\mathcal{S}^* = \{\mathbf{p} \mid |G(P(\mathbf{p}))| \leq \sin(\chi)\}$. After some algebra, we can verify that the right hand side of (42), and consequently $\dot{P}$, is non positive when $\mathbf{p} \notin \mathcal{S}^*$. Using the fact that G is decreasing, therefore invertible as well, we can rewrite the condition on $\mathcal{S}^*$ and note that it is equivalent to $\mathcal{S}$. Thus, $\mathbf{p} \notin \mathcal{S} \implies \dot{P} \leq 0$ and consequently $\mathcal{S}$ is positive invariant. ■

In conclusion, Theorem 2 states that when bounded uncertainties u_θ and u_z are present, the function P , which encodes the distance from the curve $\mathcal{C}$, admits an ultimate bound.

Note that in absence of uncertainty we have $\gamma = 0$ and $\zeta = 0$. Therefore, from $B = (\cos(\gamma) - \zeta)/(1 + \zeta)$ we have $B = 1$. Since $B = 1$, then $\sin(\chi) = 0$ and the invariant set $\mathcal{S}$ collapses to the equilibrium $P(\mathbf{p}) = 0$, which occurs only if $\mathbf{p} \in \mathcal{C}$. This formalizes the statement made in Subsection IV-A, that is, getting oriented with the field implies in convergence to the target curve.

V. TREATMENT OF FIELD SINGULARITY

In order to compute the control law (13), it is necessary to perform operations, gradients of field's components, over a normalized vector field. This normalization necessarily creates a discontinuity point in the interior of a closed curve, direct consequence of the Brouwer fixed-point Theorem [16]. The values of the field derivatives go to infinity as we approach the singularity point, which generates extremely high $\omega(t)$ commands, making it impossible to respect (17). In this section it is presented a strategy to treat this problem. It is formally defined in $\mathbb{R}^2$, but will be extended to $\mathbb{R}^3$. Without loss of generality, let $\mathbf{p}_s = \mathbf{0}$ be the singularity point. Consider a ball of radius ρ around the singularity, that is, $\mathcal{B}_\rho = \{\mathbf{p} \in \mathbb{R}^2 : \|\mathbf{p} - \mathbf{p}_s\|_2 < \rho\}$. The strategy consists in the establishment of $\omega(t) = 0$ inside the ball. In this way, while the UAV is in $\mathcal{B}_\rho$ it will course a rectilinear trajectory.

Let us define some variables based on Figure 2. The circle represents $\mathcal{B}_\rho$. Let $\beta \in \mathbb{S}^1$ be an angular coordinate to the border of the ball $\mathcal{B}_\rho$ defined as $\bar{\mathcal{B}}_\rho$. Consider also the function $\phi(\beta) : \mathbb{S}^1 \rightarrow \mathbb{S}^1$ that associates to each local β in $\bar{\mathcal{B}}_\rho$ a value $\phi(\beta) \in \mathbb{S}^1$. This function is equivalent to the orientation θ_f at $\bar{\mathcal{B}}_\rho$. We also define the function $\delta(\beta) : \mathbb{S}^1 \rightarrow \mathbb{S}^1$, in which $\delta(\beta) = \phi(\beta) - \beta$. Let $\beta_1, \beta_2 \in \mathbb{S}^1$ be coordinates relative to the intersections of a straight line path of the UAV with $\bar{\mathcal{B}}_\rho$. This path is parallel to the vector $[\cos(\theta) \sin(\theta)]^T$, and the intersections occur at instants t_1 and $t_2 > t_1$.

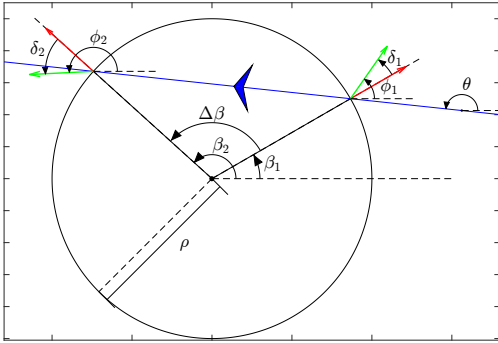


Fig. 2. Definitions for Theorem 3. In blue is the straight line with orientation θ . In red are vectors with orientations β_1 and β_2 . In green are vectors with orientations $\phi(\beta_1)$ and $\phi(\beta_2)$.

Evidently, it is not possible to ensure that the Lyapunov function V_θ in (8) will decrease while the UAV is inside $\mathcal{B}_\rho$. However, Theorem 3 establishes sufficient conditions for the Lyapunov function in the instant t_2 be smaller than in t_1 .

Theorem 3: Consider that $V_\theta(t_i)$ is the evaluation of the function $1 - \cos(\phi(\beta_i) - \theta)$ at the intersections of the straight line with $\bar{\mathcal{B}}_\rho$ at $\beta_i, i = 1, 2$. If

$$\frac{d^* \phi}{d\beta}(\beta) > 0, \quad \cos(\delta(\beta)) > 0, \quad \forall \beta \in \mathbb{S}^1, \quad (43)$$

then

$$V_\theta(t_2) < V_\theta(t_1), \quad \forall \beta_1, \beta_2 \in \mathbb{S}^1, \quad (44)$$

in which the derivative in (43) is defined as

$$\frac{d^* \phi}{d\beta}(\beta) = \lim_{\Delta\beta \rightarrow 0} \frac{\sin(\phi(\beta + \Delta\beta) - \phi(\beta))}{\Delta\beta}. \quad (45)$$

Proof: First, it is important to mention that the definition in (45) is only a formalization of the derivative of $\phi(\beta)$ in respect to β . Remember that $\sin(x) \rightarrow x$ as $x \rightarrow 0$.

Now, let $\mathcal{I}$ be the improvement function defined as $\mathcal{I} = V_\theta(t_1) - V_\theta(t_2)$, in other words, how much is the decrease of V_θ . Then, we have $\mathcal{I} = \cos(\phi(\beta + \Delta\beta) - \theta) - \cos(\phi(\beta) - \theta)$, in which $\beta = \beta_1$ and $\beta + \Delta\beta = \beta_2$.

In order to simplify the notation consider: $\phi(\beta_i) = \phi_i$ and $\delta(\beta_i) = \delta_i, i = 1, 2$. If we write the improvement function in terms of $\delta(\beta)$ we have:

$$\cos(\delta_2 + \beta_1 + \Delta\beta - \theta) - \cos(\delta_1 + \beta_1 - \theta) > 0. \quad (46)$$

Note that β_2 was replaced by $\beta_1 + \Delta\beta$. Assume without loss of generality that $\beta_2 > \beta_1$. Then, from Figure 2, trigonometric analysis allows the inference of

$$\beta_1 - \theta = -(\Delta\beta + \pi)/2. \quad (47)$$

Replacing (47) into (46), and using some trigonometric properties we find

$$-\sin\left(-\delta_2 - \frac{\Delta\beta}{2}\right) - \sin\left(\delta_1 - \frac{\Delta\beta}{2}\right) > 0. \quad (48)$$

Using the facts $\delta(\beta) = \phi(\beta) - \beta$, $\Delta\beta = \beta_2 - \beta_1$, and again after some trigonometric manipulations we have (note that the inequality was divided by 2)

$$\cos\left(\frac{\delta_1 + \delta_2}{2}\right) \sin\left(\frac{\phi_2 - \phi_1}{2}\right) > 0. \quad (49)$$

Condition in (49) is a global condition for having $\mathcal{I} > 0$. Now we need to prove that (43) imply (49). From the second condition in (43) we have that $-\pi/2 < \delta_i < \pi/2, i = 1, 2$. Then it is clear that the cosine in (49) is positive. It is then enough to show that the sine is also positive. Rewriting ϕ_2 as an integral from β_1 until β_2 we have that

$$\sin\left(\frac{1}{2} \left(\int_{\beta_1}^{\beta_2} \frac{d^* \phi}{d\beta}(\beta') d\beta' \right)\right) > 0. \quad (50)$$

If $\beta_2 = \beta_1$ the integral in (50) is zero. For any fix value of β_1 and according to the first condition in (43) the integral is a crescent function of $\Delta\beta$. The second condition in (43) implies:

$$\int_{\beta_1}^{\beta_1 + 2\pi} \frac{d^* \phi}{d\beta}(\beta') d\beta' = 2\pi. \quad (51)$$

Note that without the second condition in (43) the integral in (51) could have any value of $k \cdot 2\pi$ for $k \in \mathbb{N}$. Its value is 2π because any other case would require the violation of (43). Now, since we assume $\beta_2 > \beta_1$, we need to analyze the continuous interval of $\Delta\beta$ such that $0 \leq \Delta\beta \leq 2\pi$. The fact that the integral in (50) lies between 0 and 2π for $0 < \Delta\beta < 2\pi$ implies that the argument of the sine function in (50) lies between 0 and π and the sine is always positive.

In conclusion, we have both the cosine and the sine functions positive in equation (49), which is the global condition for $V_\theta(t_2) < V_\theta(t_1)$. ■

Theorem 3 is directly applicable only for fields in $\mathbb{R}^2$ space. In our experiments the normalized and projected fields

admitted singularities at the z axis. Then we set $\omega = 0$ and $V_z = 0$ inside a cylinder of radius ρ . In this case, we can ensure (44) if (43), computed from the normalized and projected field, is respected for all z .

VI. RESULTS

To illustrate the operation of the developed controllers we have performed simulations with the realistic model of a UAV presented in Chapter 2 of [17] with the parameters of the Aerosonde UAV in Appendix E.2 of [18]. The aircraft model has 6 degrees of freedom, 12 states and atmospheric disturbances are incorporated. Random additive disturbances (measurement and actuation) were added in order to validate the robustness of our approach. We used the same low level controllers of [13]. The results will be shown separately for 2 and 3 dimensions.

Regarding to the model in (1), we based on [13] to consider a UAV with: $v^{min} = 18m/s$, $v^{max} = 28m/s$, $v_z^{max} = 3m/s$, $\tau_z = 20s$, $\tau_\theta = 28s$ and $\tau_v = 20s$.

A. Simulations in two dimensions

The developed controller was used to make a UAV traverse a curve defined by $\alpha(x, y) = ax^4 + bx^2y^2 + cy^4 - 1 = 0$. We considered $a = 6$, $b = 0$, $c = 18$, for x and y in km . We also defined: $P = (1/2)\alpha^2$, and $G = -(2/\pi)\text{atan}(2\sqrt{P})$. The height was set to $200m$. The disturbances on the measured θ and actuated θ_c were band limited values up to $0.56rad$. In order to account for these disturbances and the model error, we set $U_\theta = 0.06rad/s$. Then, we set $k_p = 0.18$. U_v was set to $0.3m/s$. The radius of the ball $\mathcal{B}_\rho$ was set to $\rho = 0.2km$ and was enough to validate the condition of Theorem 3 and to validate the restriction on (17).

In Figure 3 it is possible to see the normalized vector field and also the trajectory executed by the UAV with initial condition $[x \ y]^T = [-0.3 \ 0]^T km$ and $\theta = \pi/6$. The set point speed was $v_r = 23m/s$. Note that the aircraft passed through $\mathcal{B}_\rho$ and then converged to the path in black. The invariant set $\mathcal{S}$, defined in Section IV-D is the region between the blue curves. In Figure 4 we can see the decay of the

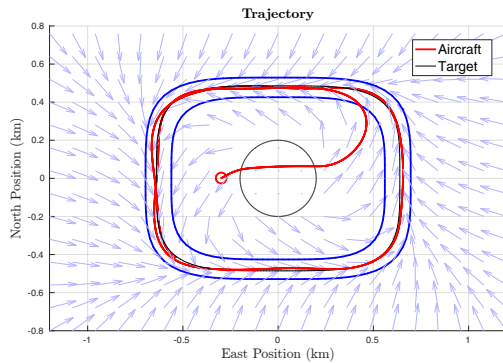


Fig. 3. Numerical simulation in $\mathbb{R}^2$. The UAV's trajectory, in red, converges to a generic path and remains inside the bounds, represented by the blue curves.

Lyapunov function V_θ . The blue/yellow dotted stretch is related to the time the UAV was traversing the $\mathcal{B}_\rho$. At this time V_θ increases but the overall effect is $\Delta V_\theta < 0$. In the

bottom we can see that the field's original potential function P increases at the beginning when the UAV is not oriented but as V_θ decay P also decay, in agreement with Subsection IV-D. The dotted red lines in Figure 4 are the correspondent ultimate bounds. In Figure 5 we can see the evolution of, from top to bottom, forward speed and bank angle. The speed stabilized at $23m/s$. Despite not considered here, the low level controllers were able to track the desired $\omega(t)$ without reaching the safe limits for the bank angle.

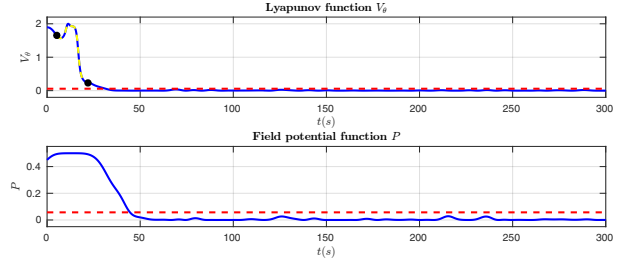


Fig. 4. In the top is the Lyapunov function V_θ . The stretch in dotted blue/yellow is associated with the time the UAV was inside $\mathcal{B}_\rho$. The dotted line is the ultimate bound. In the bottom is the field potential function P .

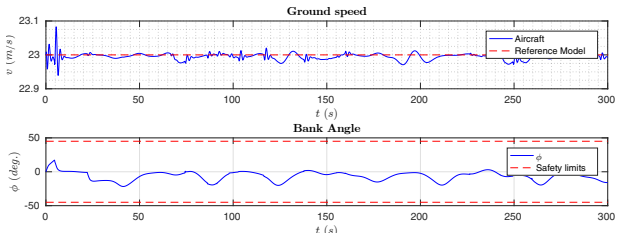


Fig. 5. Performance of the UAV in the simulation of Figure 3. From top to bottom: ground speed $v(t)$; and bank angle.

In Figure 6 we present a comparison of the trajectories resulted from the application of the proposed controllers with the trajectories resulted from the method in [13]. A circle with radius of $500m$ was considered and no additional disturbance was added to the realistic model. One can see the improvements on the paths obtained by our current approach, even when it is applied to a traverse a circle.

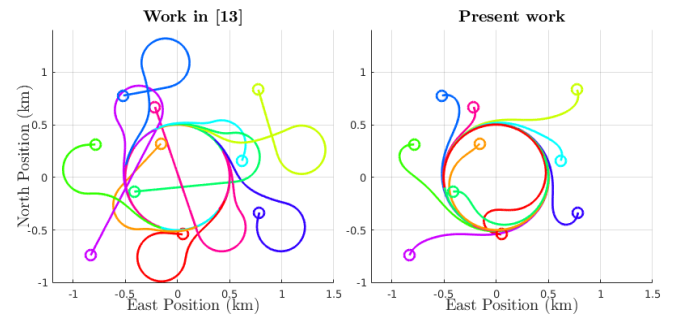


Fig. 6. Comparison of our method with the one in [13].

B. Simulations in three dimensions

For the 3 dimensional simulations we considered the basic functions $\alpha_1 = z + \mu(a^{-2}x^2 - 1)$ and $\alpha_2 = a^{-2}x^2 + b^{-2}y^2 - 1$. The values of the parameters were $a = 600m$, $b = 400m$ and $\mu = 50m$. Note that the value of μ is limited given v_z^{max} . Then we choose $P = (1/2)10^{-5}\alpha_1^2 + (1/2)\alpha_2^2$, and $G = -(2/\pi)\text{atan}(3\sqrt{P})$. The curve defined by α_1 and α_2 was shifted to an altitude of $250m$. We applied the same

disturbances θ and θ_c . We considered $U_\theta = 0.06 \text{ rad/s}$, $U_z = 0.3 \text{ m/s}$ and $U_v = 0.3 \text{ m/s}$. The controller gain was $k_p = 0.2$. The singularity was treated by considering a cylinder with radius $\rho = 0.2 \text{ km}$. Figure 7 shows the trajectory executed by the UAV with initial condition $[x \ y \ z]^T = [0.1 \ 0.1 \ 0.05]^T \text{ km}$ and $\theta = 0$. The set point speed was $v_r = 23 \text{ m/s}$. The invariant set $\mathcal{S}$ corresponds to the region inside the tube in transparent blue.

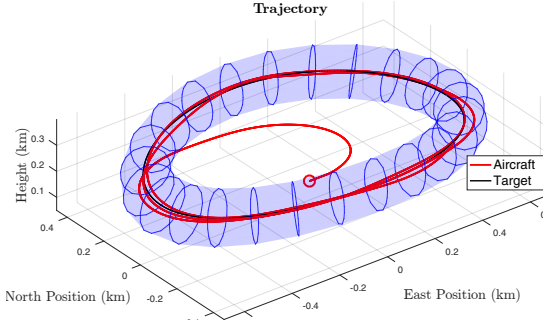


Fig. 7. Simulation considering a “saddle like” path in $\mathbb{R}^3$. In red is the trajectory followed by the UAV, which converges to the blue tube.

Figure 8 shows the decay of V_θ and P . Figure 9 shows the evolution of z and v . We can observe that the velocity v increases when the UAV is loosing height due to the coupling in the realistic model used for simulation.

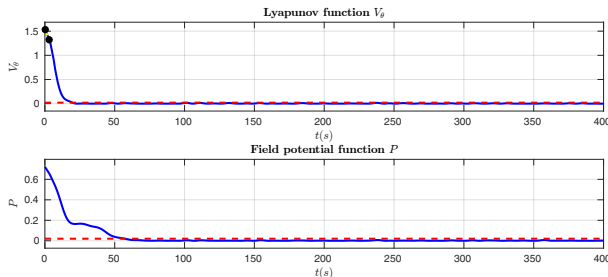


Fig. 8. Functions V_θ and P associated to the 3 dimensional simulation.

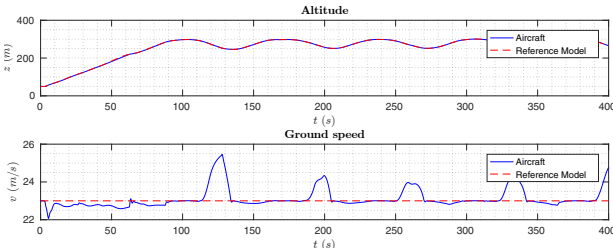


Fig. 9. Performance of the UAV in the simulation of Figure 7.

VII. CONCLUSION AND FUTURE WORKS

This work addressed the problem of controlling a fixed-wing UAV to converge to and circulate a generic closed curve that is a limit cycle of a guidance vector field. Different from most of the previous works, our controller is developed for 3D, since the computation of the feed forward parcel of the turning rate considers vertical movements of the aircraft. Furthermore, a disturbance analysis, when there are uncertainties in the model, was made and we have shown that it leads the UAV to an invariant set around the target curve as desired. Finally, singularities in the vector field were formally treated with a switch in the control law. Simulations with a

realistic aircraft model were performed in order to prove our theoretical results.

As future work we intend to develop a numerical method to compute vector fields such that all constraints imposed by our controllers, limitation of the field's derivatives and conditions of Theorem 3, are always satisfied. Finally, another extension may be the consideration of time varying vector fields.

REFERENCES

- [1] P. B. Sujit, S. Saripalli, and J. B. Sousa, “Unmanned aerial vehicle path following: A survey and analysis of algorithms for fixed-wing unmanned aerial vehicles,” *IEEE Control Systems*, vol. 34, no. 1, pp. 42–59, Feb 2014.
- [2] G. Ambrosino, M. Ariola, U. Ciniglio, F. Corrado, E. D. Lellis, and A. Pironti, “Path generation and tracking in 3-D for UAVs,” *IEEE Transactions on Control Systems Technology*, vol. 17, no. 4, pp. 980–988, July 2009.
- [3] S. Park, J. Deyst, and J. P. How, “Performance and Lyapunov stability of a nonlinear path following guidance method,” *Journal of Guidance, Control, and Dynamics*, vol. 30, no. 6, pp. 1718–1728, 2007.
- [4] D. R. Nelson, D. B. Barber, T. W. McLain, and R. W. Beard, “Vector field path following for miniature air vehicles,” *IEEE Transactions on Robotics*, vol. 23, no. 3, pp. 519–529, June 2007.
- [5] A. Alessandretti and A. P. Aguiar, “A planar path-following model predictive controller for fixed-wing unmanned aerial vehicles,” in *2017 11th International Workshop on Robot Motion and Control (RoMoCo)*, July 2017, pp. 59–64.
- [6] P. Oettershagen, A. Melzer, S. Leutenegger, K. Alexis, and R. Siegwart, “Explicit model predictive control and L1-navigation strategies for fixed-wing UAV path tracking,” in *22nd Mediterranean Conference on Control and Automation*, June 2014, pp. 1159–1165.
- [7] T. S. Andersen and R. Kristiansen, “Path-following in three dimensions using quaternions for a fixed-wing UAV,” in *2017 IEEE 26th International Symposium on Industrial Electronics (ISIE)*, June 2017, pp. 1117–1122.
- [8] E. W. Frew, D. A. Lawrence, and S. Morris, “Coordinated standoff tracking of moving targets using Lyapunov guidance vector fields,” *Journal of guidance, control, and dynamics*, vol. 31, no. 2, pp. 290–306, 2008.
- [9] D. Lawrence, E. Frew, and W. Pisano, “Lyapunov vector fields for autonomous uav flight control,” in *AIAA Guidance, Navigation and Control Conference and Exhibit*, 2007, p. 6317.
- [10] Y. A. Kapitanyuk, A. V. Proskurnikov, and M. Cao, “A guiding vector-field algorithm for path-following control of nonholonomic mobile robots,” *IEEE Transactions on Control Systems Technology*, vol. PP, no. 99, pp. 1–14, 2017.
- [11] Y. Liang, Y. Jia, J. Du, and J. Zhang, “Vector field guidance for three-dimensional curved path following with fixed-wing UAVs,” in *2015 American Control Conference (ACC)*, July 2015, pp. 1187–1192.
- [12] T. A. Jesus, L. C. d. A. Pimenta, L. A. B. Tôrres, and E. M. A. M. Mendes, “On the coordination of constrained fixed-wing unmanned aerial vehicles,” *Journal of Control, Automation and Electrical Systems*, vol. 24, no. 5, pp. 585–600, 2013.
- [13] J. L. G. Olavo, T. A. Jesus, L. C. A. Pimenta, G. D. Thums, L. A. B. Torres, and R. M. Palhares, “Robust guidance strategy for target circulation by controlled UAV,” *IEEE Transactions on Aerospace and Electronic Systems*, vol. PP, no. 99, pp. 1–1, 2018.
- [14] E. W. Frew and D. Lawrence, “Tracking expanding star curves using guidance vector fields,” in *2012 American Control Conference (ACC)*, June 2012, pp. 1749–1754.
- [15] V. M. Gonçalves, L. C. A. Pimenta, C. A. Maia, B. C. O. Dutra, and G. A. S. Pereira, “Vector fields for robot navigation along time-varying curves in n-dimensions,” *IEEE Transactions on Robotics*, vol. 26, no. 4, pp. 647–659, Aug 2010.
- [16] M. A. Khamsi and W. A. Kirk, *An introduction to metric spaces and fixed point theory*. John Wiley & Sons, 2011, vol. 53.
- [17] B. L. Stevens, F. L. Lewis, and E. N. Johnson, *Aircraft control and simulation: dynamics, controls design, and autonomous systems*. John Wiley & Sons, 2015.
- [18] R. W. Beard and T. W. McLain, *Small unmanned aircraft: Theory and practice*. Princeton University Press, 2012.